International Journal of Hospitality Management 123 (2024) 103930



Contents lists available at ScienceDirect

International Journal of Hospitality Management

journal homepage: www.elsevier.com/locate/ijhm



Full length article



The eyes have it: How do gender cues in wine labels influence U.S. women wine consumers?

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ARTICLE INFO

Keywords:
Wine
Women
Gender schema theory
Gender marketing
Marketing communications
Labels

ABSTRACT

Little is known about women wine consumers' reactions to gender cues in marketing communication despite their increasing purchasing power. This research examines the effects of gender cues (feminine vs. masculine) in wine labels on women wine consumers' attitudes, sensory expectations, sensory evaluations, and purchase intentions. Two online experiments and one field test involving 462 valid responses from U.S. women consumers reveal that they hold more favorable attitudes and intentions towards feminine (vs. masculine) wine labels. The moderating effects of in-group gender identification and wine knowledge were discussed. This study responds to calls for more research about women consumers in the hospitality field and extends the application of gender schema theory to hospitality literature. The findings offer updated knowledge on how to approach gender in hospitality marketing, particularly wine marketing. This research enriches hospitality literature in terms of gender marketing and provides substantial practical implications for the wine industry.

1. Introduction

The wine industry has traditionally been male dominated with 82 % of U.S. winemakers being male as of 2021 (Zippia, 2023), leading to a male perspective in strategizing wines (Alonso et al., 2021). Meanwhile, women drinkers have become an increasingly important market segment; more than 59 % of wine drinkers in the U.S. were women in 2021 (Thach, 2022). However, the dominance of male influence in the wine industry persists (Livat and Jaffré, 2022). For example, for decades, wine labels were rather plain, with large font names and sometimes an illustration of a castle or vineyard. This results in a lag in appealing to women consumers and may hurt wine sales, considering that women depend on wine labels more than men when unsure about making a wine selection (Atkin et al., 2007).

As tasting before consumption is precluded in most settings, consumers' expectations of the hospitality experience and products like wine primarily rely on extrinsic cues, such as labels and images (Jaud and Melnyk, 2020). Wine labels are the first cue of the wine's taste, flavor, and texture (Becher, 2015). Images presented on a bottle tend to gain the most attention and elicit more robust engagement than other elements (Jaud and Melnyk, 2020; Lunardo and Rickard, 2020). Nearly 80 % of consumers said their wine purchases are based mainly on wine labels such as color and illustration (Thach, 2019). To pique interest and win consumers' hearts, wine businesses invest heavily in label design (Celhay et al., 2020). In recent years, many wineries have started to move away from the tradition of distinguished, stately labels to colorful, trending labels that incorporate various elements (Lunardo and Rickard, 2020). Some brands include gender cues in their labels in the hope of attracting target markets (Lieven et al.,2015). For instance, emBRAZEN features on its labels historic women pioneers who defied convention and sought to bring positive changes in the world (emBRAZEN Wines, 2018). Despite the merits of gendered label design, little is known about how women consumers may react to gender cues in wine labels in the new age.

Gender marketing has been a ubiquitous trend despite the public controversies it generates (Borau and Bonnefon, 2020). Gender marketing presumes that men and women may prefer different attributes of the same product. The assumption of gender marketing is associated with social norms, gender stereotypes, and gender identity, which have undergone dramatic changes in the past decades (Fine and Rush, 2018). For instance, women consumers nowadays prefer designs that embrace their gender identity; they believe that feminine designs are not

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https://doi.org/10.1016/j.ijhm.2024.103930
Received 31 October 2023; Received in revised form 23 May 2024; Accepted 14 September 2024
Available online 18 September 2024
0278-4319/Published by Elsevier Ltd.

inherently less valuable than masculine ones (Grady, 2018). Gender identity, as a core component of self-identity (Ramkissoon and Nunkoo, 2012), has been significantly linked to various consumer variables ranging from shopping behavior to leisure activities (Palan, 2001). Many businesses have become aware of trends regarding gender identity and strategized their marketing communications to adapt to the changed gender diagram. Some businesses try to navigate a delicate line by avoiding gender-related elements (James, 2022). However, it is noted that as more consumers than ever are comfortable expressing their self-identities, gender expression should not be devoid in a product (James, 2022) because gender identity is performatively constituted through gender expression (Ulrich & Tissier-Desbordes, 2017). It is thus crucial for businesses to understand their target audience while introducing gendered marketing.

Nevertheless, inadequate attention has been paid to how gender cues may influence consumers' attitudes toward a product (e.g., Felix et al., 2022), and even fewer studies focus on women consumers (e.g., Åkestam et al., 2017), especially in the hospitality industry. Therefore, this study attempts to address the gap in the literature by examining how women consumers perceive and respond to gendered marketing efforts regarding wines, grounded in gender schema theory.

Gender schema theory is a social cognitive theory, that construes that individuals' information processing is highly influenced by their perceived gender schema (Bem, 1981). Based on the gender schema theory, we propose that gender cues in wine labels prompt women consumers to undergo a schematic process through a gendered lens, which shapes their attitudes and behaviors toward the wines. Gender schema theory has been applied to explain the effect of gender identity, largely in social/personality psychology and general psychology (Starr & Zurbriggen, 2016), but it's less commonly utilized to explain information processing in consumer studies. This study is amongst the first attempts to adopt gender schema theory in hospitality research to offer unique and new insights related to gender marketing, which, in turn, could enrich and extend the original theory.

The paper examines how gender cues in wine labels affect women consumers' purchase intentions through sensory expectations, sensory evaluation, and attitudes toward wine labels across three experimental studies. Moderating effects of gender in-group identification and wine knowledge were tested to offer contextual information. Two online experiments and one field test were conducted to test the proposed relationships. This study aims to enrich the literature on gender marketing and provide practical implications on how to approach gender in marketing for a traditionally male-dominated industry. The findings of the research should be of interest to hospitality marketing managers, restaurants, and wine business professionals.

2. Literature review

2.1. Wine labels and marketing communications

Wine labels are powerful tools in wine marketing communications (Laeng et al., 2016). As innovation in the wine industry has increasingly focused on packaging and labeling (Barbierato et al., 2023; Lunardo and Rickard, 2020), it is opportune that academic attention gear toward labels and marketing messages. Wine consumers are highly dependent upon wine labels to make purchase decisions (Escandon-Barbosa and Rialp-Criado, 2019). Previous studies have indicated that expert and non-expert wine consumers tend to focus on different attributes of a wine label (Mueller and Szolnoki, 2010). One group may find the informational content on labels important, while the other may pay more attention to the aesthetic appeal. Non-expert and casual wine consumers tend to infer clues from visual illustration and graphic design of wine labels, which in turn influence their purchase intention and taste expectations (Celhay and Remaud, 2018; Jaud and Melnyk, 2020). Ultimately, marketing effectiveness lies in consumers' interests and ability to understand the information and cues presented on wine labels (Mueller et al., 2010).

2.2. Gender identity and gender schema theory

Researchers have acknowledged that gender is an essential determinant of consumer behavior (e.g., Palan, 2001) and continues to be one of the most critical segmentation criteria used in marketing research (e.g., Kanwal et al., 2021). In wine studies, it's found that gender may influence consumers' preference of wine closure (Agnoli and Outreville, 2020), wine promotion messages (Kim et al., 2019), reciprocal behavior (Kolyesnikova et al., 2009), buying frequency (Rodríguez-Donate et al., 2020), and drinking occasions (Thach, 2012).

Gender identity, as an essential psychological attribute of one's self-identity (Palan, 2001), distinguishes males from females based on the characteristics associated with masculinity and femininity (Bem, 1981). Using gender identity as the central construct, gender schema theory posits a mental structure that enables individuals to assimilate the incoming information based on gender schema. Since proposed by Bem (1981), gender schema theory has been applied widely in psychology field to explain how individuals process gendered information (Starr and Zurbriggen, 2017) and is used to understand consumers' attitudes and behaviors in recent years (Pohlmann and Chen, 2020).

However, research support for the influence of gender schema on consumer attitude and behavior has been mixed and inconclusive (Eisend and Rößner, 2022). For instance, Lee and Lee, 2016 find that feminine males and females respond more positively to models of their own sex. In contrast, androgynous males and females respond most positively to models of the opposite sex. Gramazio et al. (2021) suggest that women's sexual portrayal in advertisements negatively affects their attitudes toward a brand. It is noted that most studies find significant gender schematic processing when gender identity is primed (Eisend and Rößner, 2022). In addition, the effects vary depending on portrayals of women and product categories, especially in critical times of sociocultural changes regarding gender roles (Tripathi et al., 2022). As more women embrace their gender identity and appreciate the feminine designs that reflect their gender identity (Grady, 2018; James, 2022), applying gender schema theory to understand women consumers in this new age can provide meaningful insights and enrich the existing literature.

2.3. Gender cues and marketing communications

There is no universally accepted definition of gender cues. Therefore, this research adopts an approach grounded in the existing literature on gender stereotypes and brand perceptions (Banaji and Hardin, 1996; Cuddy et al., 2008). First, feminine and masculine cues are typically linked to gender-specific colors or shapes; for instance, round shapes and pink colors are commonly associated with females, while angular shapes and blue colors are linked to males (Lieven et al., 2015). In addition, cues like certain symbols are often found to be associated with femininity and masculinity. For instance, dogs are related to greater ratings of masculinity (Martin, 2023). Second, gender cues can spill over to physical appearances (e.g., long hair) and traits (e.g., kindness, warmth) (Banaji and Hardin, 1996). People tend to associate femininity with concepts such as nature, body, instinct, and emotion, while masculinity is associated with logic, discipline, and rationality (Stevens and Ostberg, 2020). Women are often expected to be kind, sympathetic, warm, expressive, and caring, whereas men are often viewed as forceful, decisive, practical, and competent (Hess and Melnyk, 2016; Judd et al., 2005). The stereotypical expectations about gender often result in commonplace association with marketing cues. For instance, flowers are widely viewed as "girly" or "womanly" for their soft and delicate nature, which signal feminine traits. Floral cues are thus commonly used in marketing communications toward women consumers in an implicit manner (Perugia and Lisy, 2023).

Following the school of thoughts, the current research defines gender

cues as elements that activate individuals' associations with their perceptions of women and men. It is important to note that gender cues can be incongruent with the target sex (Lieven et al., 2015). This research focuses specifically on the effects of gender cues within groups of women, rather than comparing these effects between women and men groups.

2.4. Effects of gender cues on consumer behavior & consumer attitude

Despite the widespread use of gender cues in packaging and advertising, the conditions under which they are effective remain largely unexplored, with a few exceptions. For example, Felix et al. (2022) discover that package color induces perceptions of products' masculinity and femininity, which, in turn, influences customers' purchase intentions of green products. Pogacar et al. (2021) infer that linguistically feminine names increase positive brand outcomes. However, the feminine brand name advantage is attenuated when the typical user is male, and the products are utilitarian. Van Tilburg et al. (2015) discovered that products' physical characteristics with strong gender identification with masculinity or femininity stimulate greater affective and behavioral responses.

There is growing literature studying consumer response to wine labels (e.g., Lunardo and Livat, 2016). Wine labels are found to exert positive effects on consumers' attitudes, aesthetic appreciation, and purchase intentions (Celhay et al., 2020; Pelet et al., 2020). However, research on how gender cues influence consumer behavior, particularly women, is scant in the hospitality literature, and little is known about the effect of gender cues in a wine label on women consumers. Grounded in gender schema theory, this study proposes that individuals who identify as women are more likely to have higher purchase intention of wines with feminine (vs. masculine) labels.

H_1. Feminine (vs. masculine) wine label positively affects women consumers' purchase intention.

According to gender schema theory, people process gender-relevant information and respond based on gender schemas (Bem, 1981). Individuals who are highly masculine or feminine are likely to recall gendered information in advertisements more quickly and readily (Bem, 1981; Markus et al., 1982). When gender cues in a product/brand are congruent with the consumers' identity, they are more likely to perceive a fit and have favorable attitudes toward the product (Fugate and Phillips, 2010) and brand (Neale et al., 2016). Specifically, Feiereisen et al. (2009) find that women consumers have more positive responses to marketing communications that portray a congruent gender identity to theirs. Consumers are more favorable to advertisements when endorsers are considered in-group members representing their gender identity (e.g., Jin and Phua, 2014). In addition, attitudes toward marketing messages like ads and labels have been documented as mediators in predicting consumer behaviors (e.g., Line et al., 2016). Hence, we propose that individuals are likely to have positive evaluations of the wine labels that align with their gender identities, specifically, women consumers are more likely to have positive attitudes toward feminine (vs. masculine) wine labels, which, in turn, influence their purchase intention.

H_{2a}. Women consumers' attitude toward the feminine (vs. masculine) wine label is more positive.

H_{2b}. Attitude toward the wine label mediates the positive effect of the feminine (vs. masculine) wine label on women consumers' purchase intention.

2.5. Effects of gender cues on consumer sensory expectation & sensory evaluation

Consumers employ extrinsic and intrinsic factors to infer wine quality (Sáenz-Navajas et al., 2014). Extrinsic cues are wine characteristics that are not physically connected with the wine, like labels, aesthetics, and brand (Honoré-Chedozeau et al., 2020). Intrinsic cues are directly related to wine's physical and chemical properties, such as mouthfeel, flavor, aroma, and color. Extant research has documented that product cues influence sensory expectations of food and beverage (Orlowski et al., 2022), which further affect consumer choice, hedonic liking (Fernqvist and Ekelund, 2014), and purchase intention (Giese et al., 2014).

While there is no empirical evidence indicating that consumers' perceived femininity or masculinity of wine labels are associated with their expectation; Jaud and Melnyk (2020) find that when consumers perceive that the image on a wine label speaks to the wine, they are more likely to report a better-expected taste due to the processing fluency. Similarly, women consumers may expect the sensory attributes of wine with feminine (vs. masculine) wine labels as more aligned with the female palates, leading to higher sensory expectations and purchase intention of the wine.

Extrinsic cues such as wine labels influence consumers' product evaluation, actual sensory perception, and experience, which lead to customer satisfaction and behavioral intention (e.g., Jaud and Melnyk, 2020; Lanseng and Sivertsen, 2019). Sensory evaluations, therefore, play a central role in predicting consumers' purchase intentions. Based on the discussion above, we propose that:

H_{3a}. Women consumers' sensory expectation is higher based on the feminine (vs. masculine) wine label.

H_{3b}. Sensory expectation based on the wine label mediates the positive effect of the feminine (vs. masculine) wine label on women consumers' purchase intention.

H_{4a}. Women consumers' sensory evaluation is higher based on the feminine (vs. masculine) wine label.

H_{4b}. Sensory evaluation mediates the positive effect of the feminine (vs. masculine) wine label on women consumers' purchase intention.

2.6. Moderating effect of in-group gender identification

In-group gender identification refers to how males and females identify with their respective gender groups (Becker and Wagner, 2009). Individuals can differ in the extent to which they accept or internalize society's view of the group of women in terms of their characteristics, traits, interests, and values (Fiske et al., 2002). Previous studies showed that those who identify strongly with their group are more likely to place high importance on being a group member in their self-identity, thereby having a stronger motivation to maintain a positive image of that identity (Schmader, 2002). The impact of individuals' gender identity on their preference for a portrayal congruent with their inner identity has been established in the marketing literature (Feiereisen et al., 2009). The consistency principle or congruity theory suggests that individuals are motivated toward cognitive consistency and are willing to modify perceptions, attitudes, and behaviors to achieve harmony (Osgood and Tannenbaum, 1955). The underlying logic aligns with the gender schema theory that highly schemed individuals may be more sensitive to gender cues. In the present context, individuals highly identifying with feminine gender identity are more likely to be influenced by feminine (vs. masculine) wine labels. Hence, we propose that the effects of gender cues are strengthened for individuals with high in-group gender identification.

H_5. In-group gender identification moderates the effects of gender cues on women consumers' a) attitude, b) sensory expectation, and c) purchase intention.

2.7. Moderating effect of wine knowledge

Consumer wine knowledge refers to individuals' perception of what they know about wine based on expertise and experience (Dodd et al.,

2005). Consumer knowledge not only has a recognizable effect on information processing and decision-making but also influences the processing of heuristics (Bianchi et al., 2021). Consumers' evaluation of wine differs considerably based on their levels of wine knowledge (Ferreira et al., 2022). It is often too complex for nonexpert wine drinkers to provide a precise evaluation of the quality attributes of the wine as experts (Bianchi et al., 2021). Individuals with a high level of wine knowledge are more likely to engage in cognitive processing that requires expertise and experience in wine (Dodd et al., 2005; Ferreira et al., 2022; Kustos et al., 2019). In contrast, consumers with a low level of wine knowledge are more likely to rely on extrinsic cues like imagery on wine labels to infer the taste and flavor. Hence, we propose that the effects of gender cues may be alleviated by wine knowledge.

H_6. Wine knowledge moderates the effects of gender cues on women consumers' a) attitude, b) sensory expectation, and c) purchase intention.

Fig. 1 provided an overall framework of all the proposed hypotheses in this research.

3. Research studies

The proposed hypotheses were tested across three experiments. Study 1 set the exploratory foundation by testing the main effect of gender cues in wine labels on women consumers' purchase intention (H_1). Building upon study 1, study 2 evaluated the underlying mechanisms of attitudes toward wine labels (H_2), sensory expectations (H_3), in-group gender identification (H_5), and wine knowledge (H_6) in a controlled online experiment.

A field test of wine tasting (study 3) was conducted to validate the findings in studies 1 and 2, as well as further tested the mediating effects of sensory experience.

3.1. Study 1: Gender cues and purchase intention

To the best of our knowledge, our research is the first attempt to examine the effects of gender cues on consumers' purchase intention with manipulation in wine label images. Study 1 aimed to obtain initial descriptive evidence for the effect of gender cues in wine labels on purchase intention.

3.1.1. Stimuli design and preliminary test

To eliminate bias in consumers' perceptions regarding existing wine brands, we created fictitious labels to be used as the experiment stimuli. The wine labels were carefully designed and rigorously tested. We first conducted an extensive review (e.g., Borau and Bonnefon, 2020; Felix et al., 2022; Pogacar et al., 2021; Van Tilburg et al., 2015) as well as consulted several wine experts, including a certified sommelier, about the common elements and recent trends in wine label illustrations in the market. It's found that illustrations associated with elements and expressions like female portraits, cute animals, and flowers are typically perceived as feminine cues on wine labels; illustrations associated with elements and features such as male portraits and rugged animals are usually perceived as masculine cues on wine labels; castles, wine chateaus and grapes are commonly linked to neutral cues on wine labels. We sourced our initial pool of 24 label illustrations from Canva, an online design platform (www.canva.com). These copyrighted images were evaluated by a focus group of avid women wine consumers (n = 5). Based on the feedback received, we identified the final pool of 18 images to be tested by a separate group of women consumers through an online consumer panel (n = 90). To minimize potential noise, all elements of the wine labels were held identical except for the image illustration (e.g., Barber and Almanza, 2007; Ferreira et al., 2019; Pelet et al., 2020). The wine labels were displayed in rectangle labels made of standard white matte. All the labels were designed for red blend wine made by 10 Hands (fictional brand) in the Columbia Valley region, with a harvested vintage of 2021, and presented on a typical Bordeaux-style dark bottle with a colmated cork.

Next, a preliminary test was conducted with nighty U.S. female wine consumers who were 21 years or older and not color blind through Prolific (Prolific. com). Participants were asked to review all eighteen wine labels in a random order. For each label, participants rated the perceived masculinity and femininity level on a 100-point slider scale (0 = extremely masculine, 50 = neutral, 100 = extremely feminine) along with the perceived realism of the wine label design on a 7-point Likert scale (1= very unrealistic, 7= very realistic). Three labels were selected based on their high scores in perceived masculinity (M = 22.74, SD = 18.23), neutrality (M = 51.39, SD = 15.59), and femininity (M = 89.11, SD = 15.15), respectively, and being perceived as highly realistic (M = 6.0). Precisely, the feminine wine label was manipulated by a side portrait of a woman in a yellow dress holding a rose, with her face

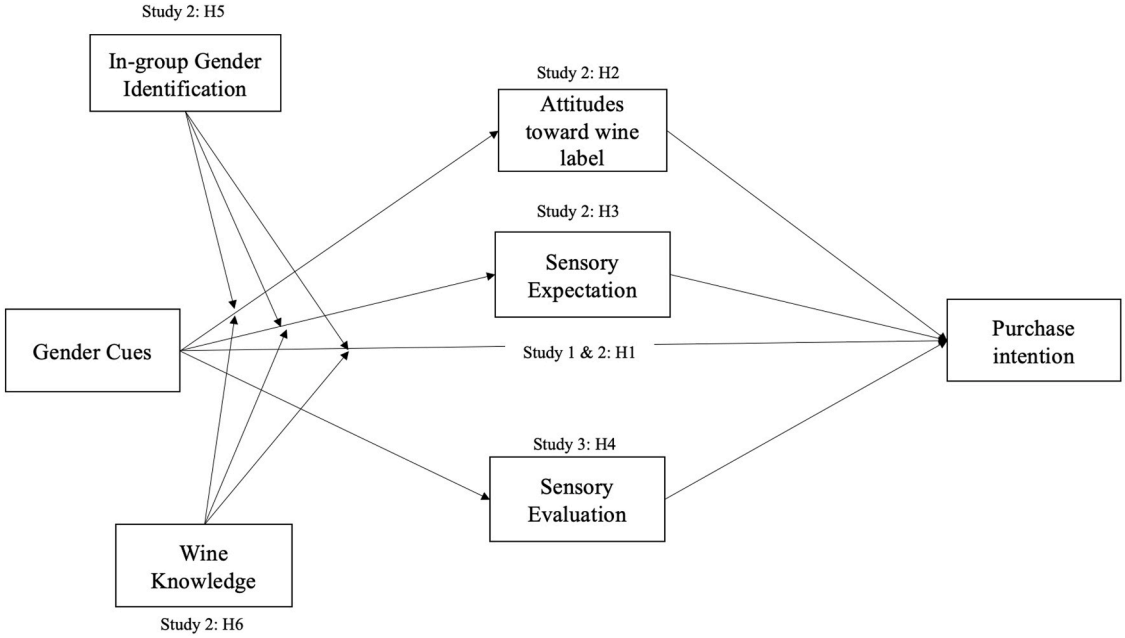


Fig. 1. Proposed conceptual model.

hidden behind the rose. The neutral wine label was illustrated with a bundle of green grapes. The masculine wine label was spoken by the face portrait of a pirate-costumed bulldog with sunglasses. Fig. 2 shows the three selected experiment stimuli.

3.1.2. Procedures and measurements

We designed a three-level (feminine, neutral, masculine) within-subject online experiment to test the main effect of gender cues in wine labels on purchase intentions (H_1). Participants were first asked to indicate their level of hunger, thirst, and fatigue as control variables, which were measured via an established 5-point scale from "1 = not at all to 5 = a great deal" following Chi et al. (2022). Participants then reviewed all three wine labels (i.e., feminine, neutral, masculine) in random order and indicated their perceived masculinity and femininity level of the label and perceived realism as in the preliminary test. Purchase intention was measured following Lu et al. (2019). Table 3 includes all the measurement items utilized in the research.

3.1.3. Data collection and sample characteristics

U.S. women consumers who were 21 or above, had consumed wine in the past six months, and were not color-blind were recruited through Prolific (prolific. com). Ninety-one qualified women consumers participated, and seventy-six responses that passed attention checks in the survey were retained for data analysis. Appendix 1 summarizes the sample characteristics.

3.1.4. Manipulation checks

Repeated ANOVA results ($F_{(2150)} = 350.33$, $p < .001$, partial $\eta^2 = .82$) indicated that participants were able to discriminate the gender cues of the wine labels as expected, with masculine cues rated as masculine ($M = 24.74$, $SD = 19.08$); neutral cues rated as neutral ($M = 53.87$, $SD = 14.59$); and feminine cues as feminine ($M = 84.71$, $SD = 15.15$), on a 100-point slider scale from extremely masculine (0) to extremely feminine (100). All three stimuli were rated as realistic (mean values of 5.36–5.92 on a 7-point Likert scale). Hence, the manipulation of gender cues was sufficient.

3.1.5. Results

Normality of the data was reviewed by skewness and kurtosis values, which were within the acceptable range for all items. None of the controlled variables (i.e. hunger, thirst, and fatigue) significantly impacted purchase intentions, which precluded them from being a potential confounding variable. Repeated measures ANOVA was employed to test H_1 to see if women consumers' purchase intentions were different upon viewing different gender-cued labels. The findings revealed statistically significant differences among the three conditions ($F_{(2150)} = 27.54$, $p < .001$, partial $\eta^2 = .27$). A follow-up pairwise differences comparison revealed that purchase intentions triggered by viewing the feminine cues ($M = 3.81$, $SD = 1.62$) were the highest, followed by neutral cues ($M = 3.52$, $SD = 1.73$), and masculine cues ($M = 2.37$, $SD = 1.39$) (see Fig. 3). Feminine cues elicited significantly higher buying intention among women consumers, compared to masculine cues ($F_{(1,75)} = 25.45$, $p < .001$, partial $\eta^2 = .25$).

3.1.6. Discussion

Findings in study 1 supported H_1 that wine labels that conveyed feminine gender cues were superior in eliciting women consumers' wine purchase intention than the ones depicting masculine cues. While study 1 established this less explored yet meaningful effect of gender cues, it remained unclear how gender cues affected women consumers' wine purchase intention.

3.2. Study 2: Underlying mechanisms

Study 2 aimed to examine underlying mechanisms that facilitated the effects of gender cues in wine labels and women consumers' behavior. Study 2 investigated the direct and mediating effects of attitudes (H_2) and sensory expectations (H_3). Furthermore, we incorporated in-group gender identification (H_5) and wine knowledge (H_6) as moderators to reveal the conditional effects.

3.2.1. Procedure and measurements

Similar procedures, like in study 1, were followed. Participants were assigned to review two wine labels (feminine and masculine) in random order and indicate their attitudes toward the wine labels, sensory expectations, and purchase intentions. Wine knowledge, in-group gender identification, and demographic questions were asked at the end. Table 3 summarized all the measurement scales adopted. Cronbach's



Fig. 2. Experiment stimuli.

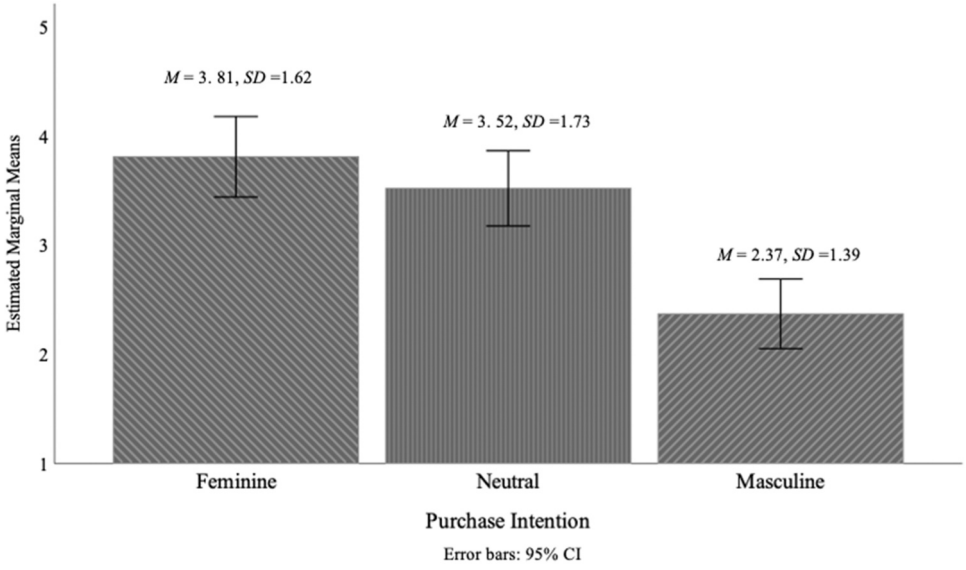


Fig. 3. Repeated ANOVA results of Study 1.

alpha values for all constructs ranged from 0.89 to 0.96, suggesting sufficient reliability. The aggregated average for each construct was used for further analysis.

3.2.2. Data collection and sample characteristics

We recruited 248 U.S. women consumers who were 21 or above, had consumed wine in the past six months, and were not color-blind via Prolific (prolific. com). Sample profiles are displayed in Appendix 1.

3.2.3. Manipulation checks and control variables

The result revealed normal distribution for all items. Repeated measure ANOVA result indicates a significant difference regarding the level of masculinity versus femininity between the masculine cue and feminine cue ($F_{(1247)} = 1109.49$, partial $\eta^2 = .82$, $p < .001$, $M_{feminine} = 82.92$, $SD = 16.93$, $M_{masculine} = 21.12$, $SD = 18.6$). Both labels were rated as realistic ($M_{feminine} = 5.38$; $SD = 1.29$; $M_{masculine} = 5.06$, $SD = 1.67$). The results suggested successful manipulation. None of the control variables (i.e., thirst, hunger, and fatigue) was found to affect the dependent variables significantly and thus were not included in the analysis.

3.2.4. Data analysis

Multiple paired-sample t-tests were used to examine the within-subject main effect of the gender cues (H_{2a} and H_{3a}). The result (see Table 1) indicated that women consumers revealed more favorable attitudes towards feminine cues than masculine cues in wine labels and reported higher liking in all sensory expectations, supporting H_{2a} and H_{3a}.

To test within-subjects mediation effects (H_{2b} and H_{3b}), we conducted the MEMORE bootstrapping procedure (model 1, bootstrap = 5,000), considering that MEMORE analysis is an apt tool in estimating the total, direct, and indirect effect of X on Y through mediators in a two-condition repeated measure design (Montoya and Hayes, 2017). The results showed significant total effect, direct effect ($\beta = .19$, *S*.E. =.07, $t = 2.76$, $p < .01$), and indirect effect ($\beta = .74$, *S*.E. =.09, 95 % *CI* [.56, .92]) of gender cues on purchase intention through attitudes toward wine labels. Thus, attitudes toward wine labels partially mediated the relationship between gender cues and purchase intention. H_{2b} was supported.

Sensory expectations were found to mediate the impact of gender cues on purchase intentions ($\beta = .84$, *S*.E. =.09, 95 % *CI* [.67, 1.01]), supporting H_{3b}. Specifically, appearance ($\beta = .39$, *S*.E. =.09, 95 % *CI* [.24,.57]) and overall liking ($\beta = .25$, *S*.E. =.10, 95 % *CI* [.07,.46]), two dimensions under sensory expectations, contributed to the significant indirect effects underlying the relationship between gendered cues and purchase intention. The output revealed a significant total effect ($\beta = .93$, $SE = .10$, $t = 8.89$, $p < .001$), an insignificant direct effect ($\beta = .09$, $SE = .08$, $t = 1.16$, $p = .25$) and a significant indirect effect ($\beta = .84$, $SE = .09$, 95 % *CI* [.67, 1.01]), suggesting a full mediation. Table 2 demonstrates the MEMORE mediation analysis results.

To examine the moderating effect of in-group gender identification (H_5), we conducted a series of MEMORE Model 2 using attitude towards wine labels, sensory expectation, and purchase intentions as outcome variables. The results suggested that in-group gender identification

Table 1
Main effect results of Study 2.

Paired labels		Attitude	Sensory expectation				
			Color	Aroma	Taste	Aftertaste	Overall
Feminine	*M*	4.37	5.03	4.77	4.74	4.56	4.78
	SD	1.60	1.49	1.23	1.32	1.28	1.41
Masculine	*M*	3.08	3.73	3.88	3.81	3.74	3.73
	SD	1.65	1.69	1.34	1.43	1.43	1.52
95 % *CI Lower*		0.92	1.03	0.70	0.72	0.63	0.83
95 % *CI Upper*		1.46	1.56	1.09	1.13	1.03	1.27
	t	8.65**	9.72**	9.03**	8.95**	8.19**	9.34**

Notes:
* $p < .05$; Sensory expectations were measured by liking of the sensory elements; $N = 248$
** $p < .005$

Table 2
MEMORE mediation results of Study 2.

	Total effect		Direct effect		Indirect effect		Bootstrap 95 % *CI*	
	β	*SE*	*β*	SE	*β*	BootSE	Lower	Upper
DV: Purchase intention								
IV: Gender cues	.93**	.10	.19*	.07				
M1: Attitudes toward wine label					.74	.09	.56	.92
IV: Gender cues	.93**	.10	.09	.08				
M2: Sensory Expectations (SE)					.84	.09	.67	1.01
SE1 – Appearance/Color					.39	.09	.24	.57
SE2 – Aroma					.08	.09	−.10	.24
SE3 – Flavor/Taste					−.02	.11	−.25	.20
SE4 – Finish/Aftertaste					.13	.08	−.03	.29
SE5 – Overall impression/liking					.25	.10	.07	.46

Note:
* $p < .05$
** $p < .01$,

significantly moderated the effects of gender cues in the proposed relationships. Fig. 4 exhibits the simple slope comparisons of the moderation effects at the high (+1 *SD*, 6.86) vs. low gender identification level (-1*SD*, 5.02). Specifically, for women consumers with higher in-group gender identification, the impacts of the feminine (vs. masculine) cues in wine labels on attitudes toward wine label (H_{5a}) and their purchase intention (H_{5c}) were more substantial. In contrast, such effects dwindled for female participants with lower in-group gender identification.

Likewise, in-group gender identification moderated the effect of gender cues on sensory expectations (H_{5b}). Specifically, for female participants with a higher level of in-group gender identification, the influence of feminine (vs. masculine) cues in wine labels on their sensory expectations pertinent to color/appearance, aroma, flavor, aftertaste, and overall liking was stronger than the one for women consumers with a low level of in-group gender identification.

The moderating role of wine knowledge (H_6) was examined by MEMORE Model 2, and all the simple slopes were presented in Fig. 5, which compares the moderating effects of wine knowledge at the low (-1*SD*, 2.32) vs. high level (+1 *SD*, 4.79). Wine knowledge was a substantial moderator in buffering the positive effect of feminine (vs. masculine) cues on attitudes toward wine labels (H_{6a}). Specifically, for female participants with a high level of wine knowledge, the influence of feminine (vs. masculine) cues on their attitudes toward the wine label was diluted. In contrast, the influence of feminine (vs. masculine) cues on attitudes toward the wine label was stronger for female participants with a low level of wine knowledge.

The interplay between gender cues in wine labels and wine knowledge on sensory expectations was also significant (H_{6b}). The positive impact of feminine (vs. masculine) cues in wine labels on sensory expectations (i.e., aroma, flavor, and aftertaste) was more powerful for female participants who reported low wine knowledge levels. However, wine knowledge was not found to moderate the relationship between gender cues in wine labels and purchase intention (H_{6c}). The insignificant role of wine knowledge in influencing purchase decisions suggested that feminine labels demonstrated a prominent effect in swaying women consumers' purchase intention irrespective of their wine expertise.

3.2.5. Discussion

Findings in study 2 not only confirmed the results in Study 1 but also revealed the significant mediation of attitudes toward wine labels and sensory expectation in the relationships between gender cues and purchase intention. The results also indicated significant moderation of in-group gender identification on the effect of gender cues on attitudes toward wine labels, sensory expectation, and purchase intention. Specifically, the positive effect of feminine (vs. masculine) wine label was more profound for women consumers with high (vs. low) in-group gender identification. Study 2's results highlighted partial moderation of wine knowledge. The effect of feminine (vs. masculine) wine label was more salient on attitudes toward the wine label and sensory expectation for women consumers with low level of wine knowledge than those who were more knowledgeable in wine, but wine knowledge did not significantly modulate the effects of gender cues on women consumers' purchase intention.

3.3. Study 3: Sensory evaluations in the field test

Whilst the findings in Studies 1 & 2 uncovered how women consumers reacted to the gender cues conveyed in wine labels, it was unclear if it would hold true when actual tasting was involved. We designed and conducted a between-subject experiment in a dining room with a controlled tasting section in a major U.S. research university. In addition, we invited an expert panel consisting of six certified wine professionals to conduct a blind tasting to offer benchmark sensory input of the wine quality attributes.

3.3.1. Experiment procedures and data collection

Study 3 was executed in the following phases. First, we selected one of the most common and preferred wines based on the latest wine market reports (Silicon Valley Bank Wine Division, 2023) from a local winery. Specifically, it's Cabernet Sauvignon and Merlot red blend from 2019 with medium body (i.e., a moderate weight and mouthfeel), medium tannin (i.e., a noticeable but not overpowering bitterness and dryness on the palate), and medium alcohol level (i.e., around 13 % alcohol by volume). The wine sample tasting notes were provided by our expert panel. According to their assessment, this wine exhibited prominent non-fruity notes, mainly characterized by sweet spices, tobacco, and peppers, with subtle hints of black and red fruit flavors. Meanwhile, the wine is also described as having a medium body, a medium level of alcohol content, and tannins. The wines for the experiment were sourced in "shiners" (i.e., clean labeled wines). We printed wine labels and affixed them to the wine bottles to manipulate feminine and masculine cues (identical to the ones in studies 1 & 2, see Fig. 6). Therefore, the exact wine in differently labeled bottles was the experiment stimuli for study 3.

To increase sample representation, the field experiment was conducted during a homecoming week when there were many out-of-town visitors. Female customers who are 21 or older were recruited from both online (social media) and offline (on campus). Upon arrival at the dining room, participants were greeted and instructed by trained research assistants to review and sign an informed consent and then complete the first two sections of the survey on a digital device (smartphone or tablet). The first section included questions about wine knowledge and control variables like hunger, thirst, and fatigue. In the second section, participants were randomly assigned to review one of the wine labels (feminine vs. masculine) and rate their sensory expectations, perceived masculinity-femininity, and realism based on the wine label.

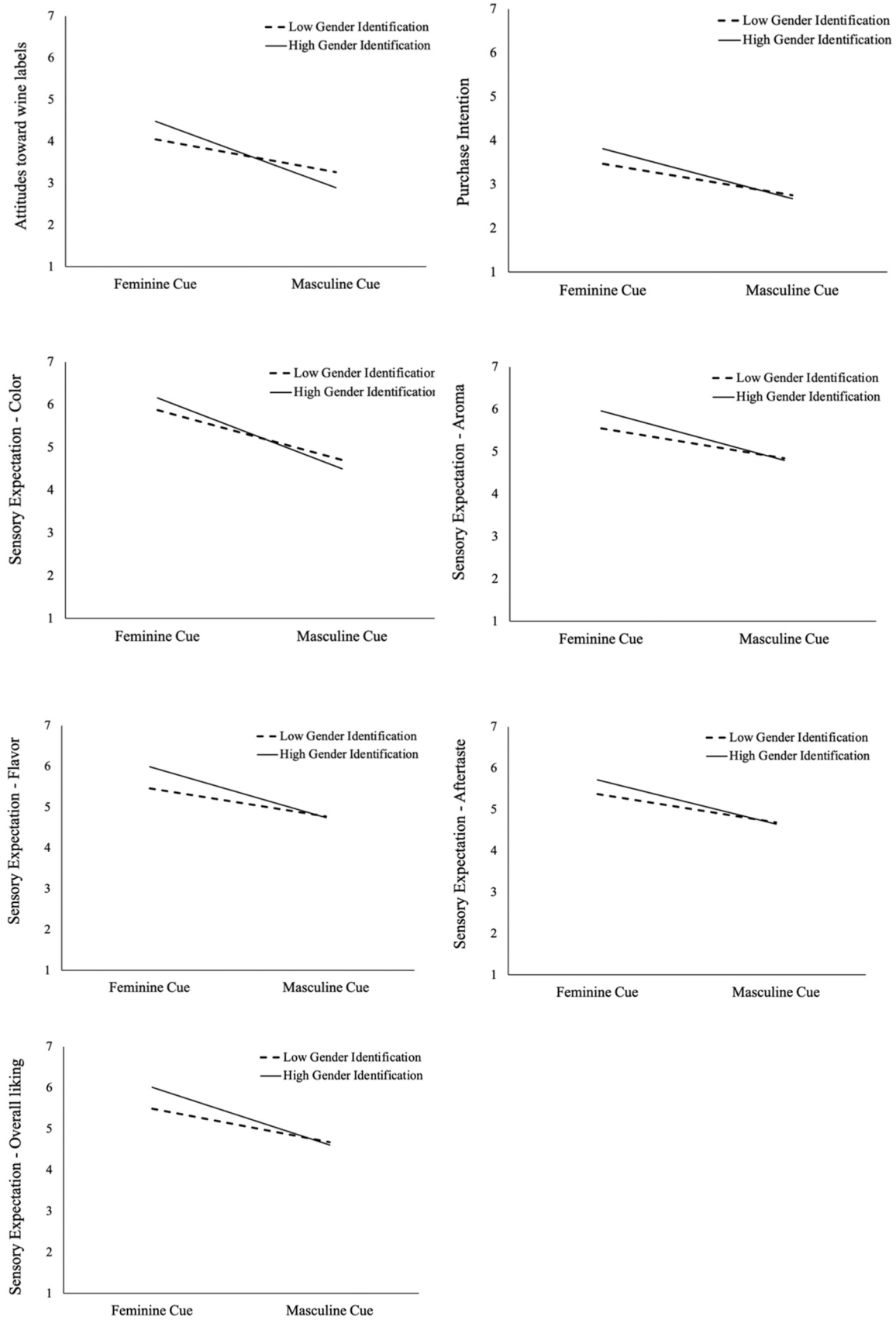


Fig. 4. Significant moderating effects of in-group gender identification in Study 2.

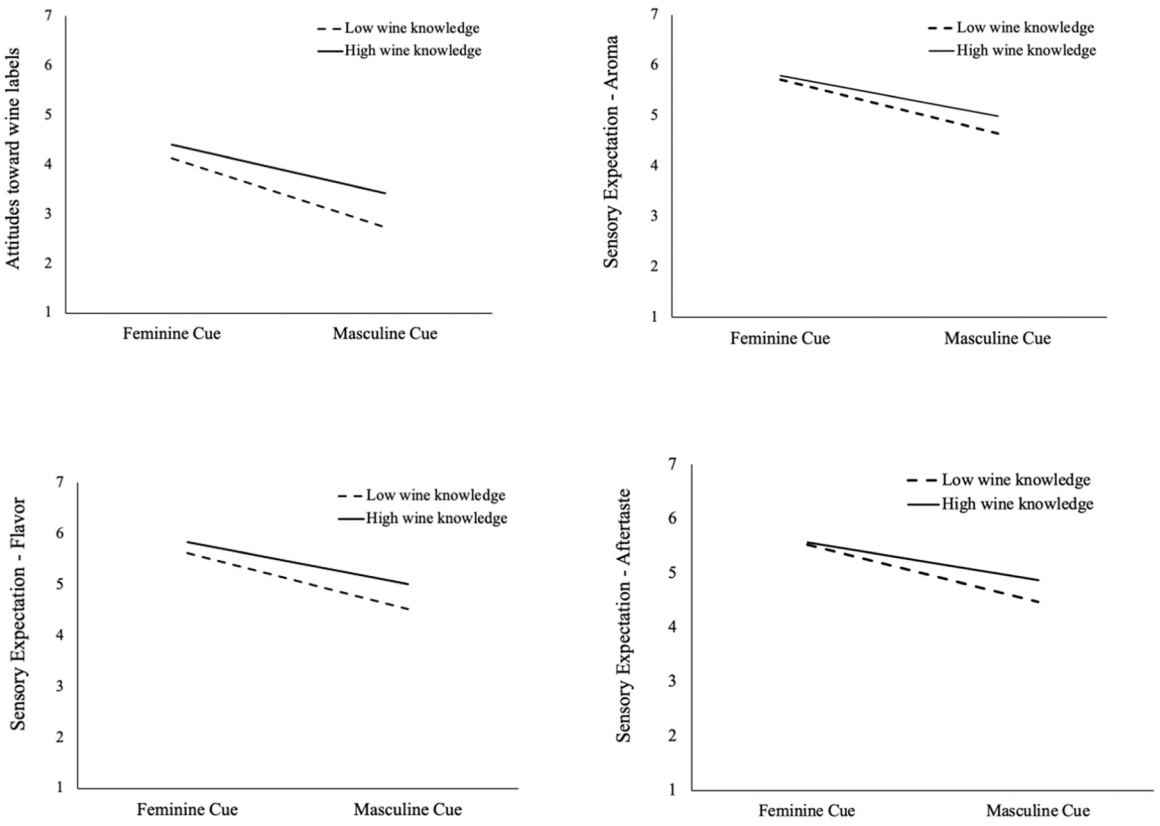


Fig. 5. Significant moderating effects of wine knowledge in Study 2.

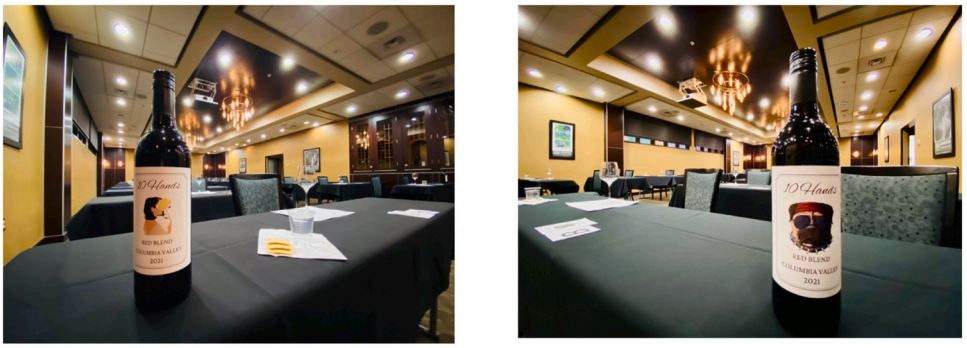


Fig. 6. Stimuli of Study 3.

Afterward, participants were escorted to the tasting section in a dining room. Each participant was seated at a solo table (furnished with water, crackers, white paper, spit bucket, etc.). Trained research assistants poured the wine from a labeled bottle and confirmed with the participants that the wine label was consistent with what they reviewed in the previous section. After tasting, participants provided sensory evaluations, purchase intention, and demographics. Upon exit, the participants were debriefed and offered a small souvenir. Fig. 7 illustrates the field experiment procedure.

3.3.2. Measurement

Most measures were the same as in Study 2. Sensory expectation and sensory evaluations were based on the liking of five core wine attributes in Study 2, i.e., color, smell/aroma, taste/flavor, aftertaste, and overall liking/impression (Harrington, 2007, p. 42). Participants were also asked to evaluate the categorized aroma and flavor (e.g., fruity, non-fruity, earth, mineral, woody), level of body/weight structure (e.g., light-bodied, medium-bodied, full-bodied) based on the deductive wine appreciation guidelines of the Court of Master Sommelier (mastersommeliers.org/resources). Table 3 summarizes the measurement used in our research.

3.3.3. Manipulation checks

A total of 153 women consumers participated, and 138 responses were useable ($N_{feminine}$=69, $N_{masculine}$=69). No control variables were found to confound the study results. All items showed normal distribution. Manipulation was effective based on the independent *t* test, $t_{(136)} = -27.50$, $p < .001$, $M_{masculine} = 23.35$, $SD = 14.43$, $M_{feminine} = 84.62$, $SD = 11.59$. Participants perceived both labels as highly realistic ($M_{feminine} = 5.68$, $SD = 1.62$;$M_{masculine} = 5.20$, $SD = 1.64$).

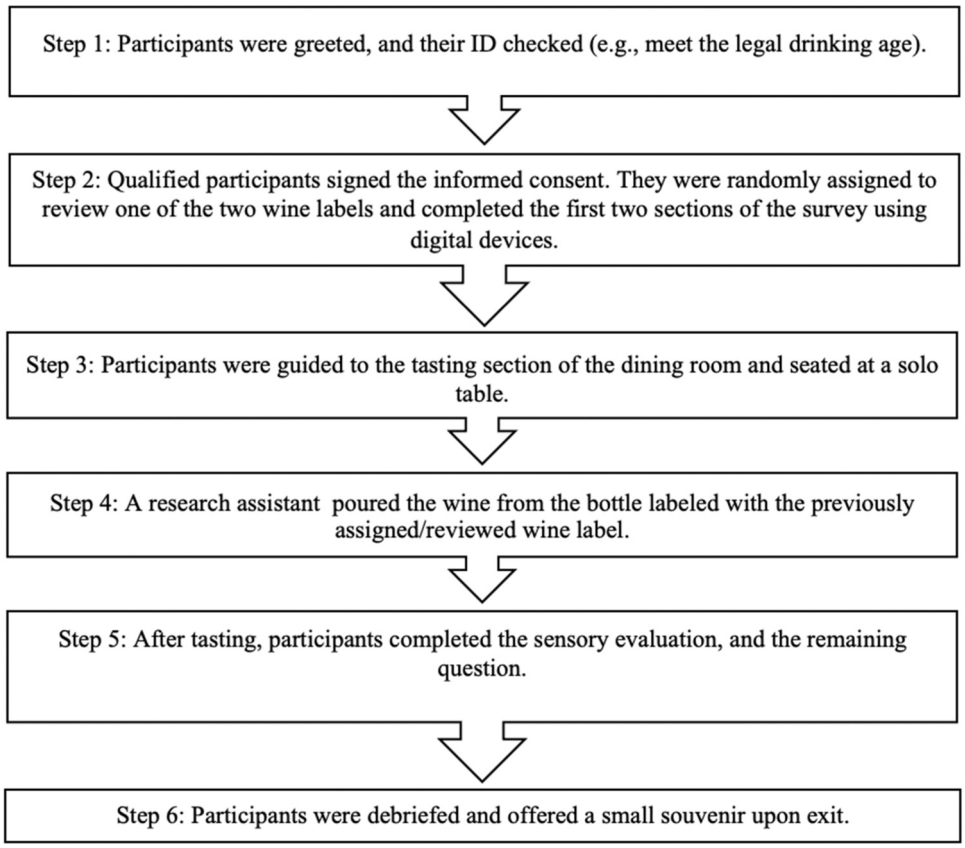


Fig. 7. Flowchart of the field experiment in Study 3.

3.3.4. Results

To investigate H_{4a}, MANCOVA was applied, controlling hunger, thirst, fatigue, wine knowledge, and all the attributes under the sensory expectation. Female customers reported a higher liking of aroma, flavor, aftertaste, and overall impression when tasting masculine-labeled wine compared to feminine-labeled wine (see Table 4). Hence, H_{4a} was not supported.

The result also suggests that female customers reported different dominant flavors when tasting different labeled wines, notwithstanding it's the same wine. Specifically, 21.73 % (N =15) of participants who tasted feminine labeled wine reported fruity flavors such as red currant, black plum, and blueberry cues, significantly more than the participants (7.24 %, N = 5) who sampled masculine labeled wine (p <.05). Furthermore, there were significantly fewer female participants reporting mineral flavors such as minerality and limestone when tasting feminine-labeled wines (18.84 %, N = 13) than masculine-labeled wines (31.88 %, N = 22), p <.05.

To test H_{4b} (the mediating effect of sensory evaluation), Haye's PROCESS model 4 was operated by controlling hunger, thirst, fatigue, knowledge, and all eight sensory expectation attributes. The liking of taste appeared to be a full mediator between gender cue and purchase intention, β =.24, 95 % *CI* [.01,.57], despite an insignificant direct relationship between gender cue and purchase intention, β = −.03, *S.E.* =.22, p =.91. Similarly, the liking of overall impression fully mediated the relationship between gender cue and purchase intention, β =.15, *CI* 95 % [.01,.42], given a significant indirect effect and an insignificant direct effect. H_{4b} was supported. Table 5 illustrates the mediation analysis results.

3.3.5. Discussion

Findings in Study 3 extended those in studies 1 & 2 by discussing the effect of sensory evaluations. The results indicated that sensory evaluations fully mediated the effect of gender cues on women consumers' purchase intention after tasting the wine. Specifically, gender cues in wine labels influenced female customers' liking of the sensory attributes and their categorization of the sensory attributes. Female customers were more likely to link feminine cues with fruity flavors, even though the same wine was served. Such inferences were associated with consumers' sensory evaluations and purchase intentions.

4. Discussions and implications

4.1. General discussion

The primary purpose of this research is to explore an unacquainted segment – female wine consumers, given their rising consumer power (Thach, 2022), and investigate how this cohort responds to a novel wine label design that highlights the feature of gender cues (feminine vs. masculine) through three sequential studies that involved data from both online experiments and field experiment, corresponding to retail and restaurant/winery settings. Table 6 summarizes the results of hypothesis testing. It's noteworthy that the respondents of this research closely mirrored U.S. wine consumers. Similar to the general wine consumer base (Silicon Valley Bank Wine Division, 2024), the age of the participants in this study spanned from 21 to 70 years old and above, with Millennials (28−43) and Baby boomers (60 – 69) being the predominant contributors to wine consumption.

Our research acknowledges the positive role of feminine (vs. masculine) wine labels in influencing women consumers' purchase intention through between- and within-subjects design. The positive effect of feminine cues in wine labels is consistent with previous findings that women consumers are inclined to respond more favorably toward products or brands when a shared identity is presented (e.g., Lee and Lee, 2016).

Table 3
Measurement items for Study 1–3.

Factors/Items	Sources
Attitudes toward wine labels	Chi et al. (2022)
I like this label a lot.	
This label is very appealing to me.	
This label makes me feel good.	
I am very interested in this label.	
Sensory expectation	Harrington (2007)
Color/Appearance	
Aroma	
Flavor	
Aftertaste	
Overall liking	
Purchase intention	Lu et al. (2019)
The probability that I will buy the wine is high	
It is likely that I will buy the wine	
I am certain that I will buy the wine.	
Wine knowledge	Dodd et al. (2005)
I am confident in my wine knowledge.	
Among my friends, I am a wine expert.	
I know more about wine than others do.	
In-group gender identification	Becker and Wagner (2009)
I identify with the group of women.	
I feel strong ties to other women	
Overall, being a woman is an important part of my self-image	
Being a woman is important to me.	
Sensory evaluation	Harrington (2007)
Color/Appearance	
Aroma	
Flavor	
Aftertaste	
Overall liking	
Aroma(Smell)	Guideline of Wine Appreciation, Court of Master Sommelier (mastersommeliers.org)
•Fruity aroma driven (e.g., red currant, black plum, blueberry)	
•Non-fruity aroma driven (e.g., floral, vegetal, herbal, spices)	
•Earth aroma driven (e.g., forest floor, compost, mushroom)	
•Mineral aroma driven (e.g., minerality, limestone)	
•Woody aroma driven (e.g., French oak, American oak)	
Flavor(Palate) – Same as Aroma categories	
Body	
•Light bodied-Medium minus-Medium – Medium plus – Full bodied	

Table 4
Main effect of gender cue on sensory evaluation in Study 3.

Label		Sensory evaluation				
		Color	Aroma	Taste	Aftertaste	Overall
Feminine	*M*	5.92	5.19	4.63	4.69	4.92
	SD	1.06	1.43	1.71	1.54	1.43
Masculine	*M*	5.97	5.36	5.19	5.07	5.48
	SD	1.07	1.30	1.67	1.71	1.42
	M^2	1.63	4.53	9.83	8.38	6.65
	F	1.51	2.86	4.63	4.10	4.31
	p	.12	<.001	<.001	<.001	<.001

Note: N=138; *df* =136; Sensory evaluation was measured by liking of each sensory element.

Moreover, findings across three studies showcase how gender cues influence women consumers' purchase intentions when they cannot taste the wine and can taste the wine (restaurant/winery setting).

Table 5
Mediation analysis results of Study 3.

	DV - Purchase Intention							
	Total effect		Direct effect		Indirect effect		Bootstrap 95 % *CI*	
	β	*SE*	*β*	SE	*β*	BootSE	Lower	Upper
IV								
Gender cues	.46	.33	−.03	.22				
Mediators								
Total indirect effects					.48	.26	.01	1.03
Liking of color					−.01	.04	−.10	.07
Liking of aroma					.01	.03	−.05	.09
Liking of flavor					.24	.14	.01	.57
Liking of aftertaste					.11	.11	−06	.32
Overall liking					.15	.11	.01	.42
Aroma					−.03	.06	−.16	.09
Flavor					.01	.04	−.06	.10
Body					.00	.03	−.07	.07

Notes: For categorical IV, feminine is coded as 0 while masculine is coded as 1.

Table 6
Summary of hypothesis testing results.

Hypothesis	Study 1 (online)	Study 2 (online)	Study 3 (field)
H_1. Gender cues → Purchase intention	√	√	-
H_{2a}. Gender cues → Attitude toward wine label	-	√	-
H_{2b}. Gender cues → Attitude → Purchase intention	-	√	-
H_{3a}. Gender cues → Sensory expectation	-	√	-
H_{3b}. Gender cues → Sensory expectation → Purchase intention	-	√	-
H_{4a}. Gender cues → Sensory evaluation	-	-	X
H_{4b}. Gender cues → Sensory evaluation → Purchase intention	-	-	√
H_5. Moderating effect of in-group gender identification on			
a) Gender cues → Attitudes toward wine label	-	√	-
b) Gender cues → Sensory expectations	-	√	-
c) Gender cues → Purchase intention	-	√	-
H_6. Moderating effect of wine knowledge on			
a) Gender cues → Attitudes toward wine label	-	√	-
b) Gender cues → Sensory expectation	-	√	-
c) Gender cues → Purchase intention	-	X	-

Note: √= Supported; - Not tested; X= Not supported

Building upon the positive effect of gender cues in study 1, study 2 & 3 unearthed attitudes toward wine labels, sensory expectations, and sensory evaluation as vital mechanisms elucidating the relationship between gender cues and purchase intention. Specifically, study 2 finds that when no tasting is involved, wines with feminine (vs. masculine) labels elicit higher visual sensory expectation (appearance) and overall impression from women consumers; and these two dimensions of sensory expectations are also essential in mediating the effect of feminine cues on purchase intention. The findings indicate that without actual tasting experience, consumers primarily rely on easily accessible heuristics (i.e., what they can see) and general impressions to evaluate the wine quality and make a judgment, as advocated by Rahman and Reynold (2015). Our results also denote Van Tilburg et al.'s (2015) conclusion on how feminine symbols generate grander behavioral responses.

When actual tasting is involved (study 3), the sensory evaluations of flavor and overall impression become dominant in mediating the

relationship between gender cues and female customers' purchase intention. The prominent role of flavor and overall impression aligns with the existing literature that emphasizes that the evaluated taste of wine is the most crucial factor in determining wine consumers' decision-making (Rahman and Reynolds, 2015). A surprising finding is that female customers indicate a lower liking of flavor and overall impression after tasting wines with feminine (vs. masculine) labels. We infer that this is caused by an incongruence between the expected flavor influenced by the feminine label and the actual flavor of the wine sample. Results in study 3 show that consumers are likely to infer the fruity-driven flavor of wines when labeled with feminine cues. Yet, our wine sample (non-fruity flavor) with a feminine label appears to be less fruity than the women consumers expected. The finding reinforced the power of gendered labels in affecting female customers' sensory experience interpretation. In addition, the findings indicate that the congruency between sensory expectation inferred by gender cues in wine label and sensory evaluations in actual tasting is important for customers' decision-making.

Moreover, the research uncovers the moderating effects of in-group gender identification and wine knowledge (Study 2). Our findings indicate that a higher level of identification with being women strengthens the positive effect of feminine cues on women consumers' attitudes towards wine labels, sensory expectations, and purchase intention. The outcomes reiterate the importance of congruence between product and women consumers' gender identification (e.g., Feiereisen et al., 2009).

Wine knowledge buffers the positive effect of feminine labels on women consumers' attitudes toward the label and sensory expectations. However, it does not attenuate the effect of gender cues on purchase intention. Knowledgeable consumers have cognitive resources available to process the information through a systematic route and thus are more likely to seek concrete clues to evaluate the wine's sensory qualities. In contrast, nonexpert women wine consumers rely more on available heuristics (e. g., feminine wine labels) (Bianchi et al., 2021). However, when it comes to making purchase decisions, women consumers across varying levels of wine knowledge are swayed by feminine wine labels. This reflects the societal trend that women consumers in general embrace products that speak to their feminine identity (Fugate and Phillips, 2010).

4.2. Theoretical implications

The research responds to the call for more attention to women in hospitality literature (Mooney, 2020; Russen et al., 2021). Previous studies regarding gender often focused on distinctions between female and male consumers (Feiereisen et al., 2009), overlooking the within-group differences. This research reveals a different angle by investigating women consumers' evaluations of gender cues at varying levels of in-group gender identification. The findings offer updated insights into gender cues in hospitality marketing communication.

The findings offer first-hand insights into gender marketing in the hospitality context. Notably, this research extends the application of gender schema theory to the wine consumption context to reveal how understudied women consumers process gender cues in hospitality products. Although gender schema theory has been used for understanding gender identity in psychology, it has rarely served as a theoretical foundation in understanding information processing in consumer studies, particularly in hospitality literature. The findings provide empirical evidence across three experiments that validate the gendered lens in processing gendered wine labels.

This paper contributes to the hospitality literature by revealing the differentiated roles of attitude, sensory expectation, and sensory evaluation. Although recent studies hinted at the effects of extrinsic cues on consumers' purchase intention in food (Orlowski et al., 2022), little is known about how extrinsic cues (e.g. gender cues) may influence consumers' sensory expectations and evaluations. The research discloses a unique perspective by looking into and delineate the positive effects of gender cues on sensory expectations and experiences among women consumers. The research reaffirms the importance of cognitive processing of sensory evaluation (e.g., Lu et al., 2019) and integrates the affective experience of attitude toward wine labels (Jaud & Melynk, 2020).

In addition, the research involves the conditional effects of in-group gender identification and wine knowledge to provide a comprehensive view of gender cues in hospitality marketing communications. The findings regarding gender identification connote the gender schema theory that individuals can have varying levels of gender identification (Bem, 1981), and their in-group identification influences their behavior. The research joins the previous effort in applying gender identity and identification rather than biological sex in explaining behaviors (e.g., Van Tilburg et al., 2015). However, findings of the boundary effects of wine knowledge are inconsistent with previous literature concerning its role in swaying wine quality evaluation and purchase decision-making for general consumers (e.g., Lu et al., 2016). This study denotes that for women wine consumers, even knowledgeable ones, are more likely to purchase wines with feminine wine labels.

4.3. Practical implications

Findings of the paper provide timely suggestions for hospitality businesses by addressing the previous call for more evidence on women consumers. This research notes that incorporating gender cues into the label design helps to entice women consumers and elicit more favorable evaluations. As many wineries are open to trying new labels, they are encouraged to experiment with innovative and diverse designs that reflect the shifts in societal trends and customize their marketing messages to women consumers. Wine businesses may consider more inclusive consumer and expert panels while designing wine labels and marketing campaigns to ensure up-to-date perspectives. It's worth pointing out that it's ultimately up to the consumers, not designers or marketers, to perceive/decide whether a label is feminine in marketing communications. It's, therefore, strongly advisable for wine businesses to pilot-test their wine labels for femininity cues among target consumers, as we did in our study. It's also essential to continuously gather insights and feedback from the target audience to refine the designs.

Our findings also indicate that gender cues in wine labels influence women consumers' sensory expectations, evaluation, and behavioral intention. Findings indicate that when the wine has the right label, favorable attitudes toward the wine label and hedonic liking of the wine sensory qualities will ensue, which further prompt women consumers' intention to purchase the wine. Our findings shed light on the stereotype implicit in wine consumption, i.e., consumers often associate fruity aroma/flavor and light-bodied wine with feminine labels, including women consumers in this study. Wine marketers should label wines to live up to such expectations to ensure a congruent experience for consumers because incongruity leads to disconfirmation of expectation and lower evaluation of the wine, as evidenced by this study.

In line with recent societal trends, findings suggest that feminine wine labels highly resonate with women consumers who embrace their women identity. Wine businesses can utilize consumer data to tailor marketing communication and apply gender cues accordingly. For instance, when targeting women consumers with high in-group gender identification, wine businesses should leverage the power of gendered cues, especially feminine cues, to elicit positive responses.

While women consumers who are highly knowledgeable about wine are less likely to be influenced by gender cues embedded in the wine labels to infer sensory expectations, feminine labels play a prominent role in affecting women consumers' purchase intention, regardless of their wine expertise. This suggests that feminine labels trump wine expertise among women consumers in making wine purchase decisions and demonstrates the powerful effect of feminine wine labels in appealing to women consumers. The finding reaffirms the critical

importance of creating marketing messages that speak to women audiences in order to win over their hearts and purse.

5. Limitations and future studies

This research bears limitations that encourage future studies. First, the paper focuses on women consumers and their reactions to gender cues in hospitality marketing due to their increasing purchasing power which has been understudied previously. Future studies are recommended to explore the perceptions of male consumers and examine how they may react to gender cues. Such a study would bring new and substantial insights into the literature and the practice. Second, the research was conducted in the U.S.; therefore, the findings may only be generalizable to regions with similar cultural and social backgrounds. Gender identity is inherently shaped by social norms and cultural influences (Bem, 1981), and people from different regions may hold different views about gender cues in marketing communications. Future studies could examine the similarities and differences in how consumers process gender cues through cultural lenses. Third, while our sample was representative of U.S. women wine consumers, it is recommended that future research conduct further segmentation by generation cohort, ethnicities, and other relevant criteria, which would enrich and enhance our understanding of subsets of women consumers. Lastly, the study used wine as the focal product to showcase the effect of gender cues and offer contextual insights into the male-dominant industry. The conclusions may only be applied to like products, and future studies may extend the research findings to different types of products to deepen the understanding of gender cues in a broader context.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We express our deepest gratitude to Hamilton Cellars (http://www.hamiltoncellars.com/) from Red Mountain, WA for supporting our research by providing all the shiners.

Appendix. – Demographic profile

		Study 1	Study 2	Study 3
		Distribution (%)		
		(*n*=76)	(*n*=248)	(*n*=138)
Age	18–24	11.8	11.2	69.5
	25–34	39.3	30.8	22.9
	35–44	19.5	22.0	3.5
	45–54	14.3	14.0	0
	55–64	7.8	13.6	0.7
	65 or over	6.5	7.6	2.1
Ethnicity	White	82.9	84.7	72.5
	Black or African American	9.2	8.1	2.2
	American Indian or Alaska Native	2.6	1.2	0.7
	Asian	6.6	8.5	23.9
	Native Hawaiian or Pacific Islander	0.0	0.4	1.4
	Other	3.9	3.6	18.8
Education	Less than high school	0.0	2.0	0.0
	High school graduate/G.E.D	21.1	25.0	16.7
	Associate degree/certificate	14.5	19.0	12.3
	Bachelor's degree	47.4	39.1	52.9
	Master's degree	13.2	13.3	8.0
	Doctorate	3.9	1.6	8.0
Annual Income	Less than $30,000	23.7	23.4	56.5
	$30,000 - $44,999	15.8	14.9	6.5
	$45,000 - $59,999	19.7	10.9	3.6
	$60,000 - $74,999	10.5	14.5	5.8
	$75,000 - $89,999	2.6	14.5	4.3
	$90,000 - $104,999	3.9	4.4	4.3
	$105000 and above	23.7	17.3	15.9
Wine purchase frequency	Never	0.0	0.8	5.8
	Less than once a month	25.0	32.3	37.0
	Once a month	28.9	32.7	23.9
	2–3 times a month	31.6	23.4	26.1
	Once a week	9.2	8.4	3.6
	2–3 times a week	5.3	2.0	1.4
	Daily	0.0	0.4	2.2
Wine drinking frequency	Never	0.0	0.0	0.0
	Less than once a month	19.7	23.0	33.3
	Once a month	11.8	16.9	13.0
	2–3 times a month	31.6	27.4	31.2
	Once a week	14.5	10.9	14.5
	2–3 times a week	18.4	17.7	5.8
	Daily	3.9	4.0	2.2
Average expenditure on a bottle of wine (US$)	Under 10	10.5	16.9	4.3
	10–20	69.7	52.7	53.4
	21–30	15.8	20.1	38.1
	31–50	1.3	8.8	2.8
	51 and above	2.6	1.2	1.4

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